

Mycroft — Chapter 1 Workbook

The Fluency Trap: Completed Exercises & Your Own Mycroft Project

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Project: **Your Own Mycroft** — Investment Research Desk

Thesis under audit: **Titan Company (India: TITAN)**

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Part 1 — Conceptual Exercises (1–9)

Warm-up

Exercise 1 — Disaggregate a fluent claim

Sample sentence: *"Operating expenses increased this quarter because we ramped up hiring in engineering."*

(a) **The claim:** OpEx increased quarter-over-quarter.

(b) **The causal language:** "because we ramped up hiring in engineering" — asserts a single cause without qualification.

(c) **Source that would verify it:** Headcount report by department for the period, cross-checked against payroll/OpEx ledger to confirm the increase is attributable to compensation expense specifically — not a one-time contractor payment or software license renewal.

Exercise 2 — First row of an assessment artifact

Opening thesis claim audited: *"Revenue is down this quarter because enterprise renewals slipped in the mid-market segment, which accounts for approximately 23% of recurring revenue."*

Claim	Source	Period	Owner	Status	Gate Required
Enterprise renewals slipped in mid-market, driving revenue down	Not identified — no renewal export referenced	"This quarter" — not specified	Not identified	Unsupported	Obtain renewal export from CRM, identify period boundary; enterprise/mid-market lead must confirm renewal timing as actual driver

Exercise 3 — Bad answers to the three supervision questions

Scope (bad answer): "Look at recent trends and explain what's going on with revenue." — no period boundary, no entity/segment, no source files named.

Approval (bad answer): "It'll go in Friday's deck if it looks good." — no named human accountable for clearing the gate; "looks good" is a fluency judgment, not adequacy.

Verification (bad answer): "It reads correctly and uses the right terminology, so it's probably fine." — confuses surface fluency with evidentiary support.

Application

Exercise 4 — Full assessment artifact for a monthly close commentary

Sample commentary audited: *"Close for October was clean overall. Cash increased 4% driven by faster collections. AP aging improved slightly, though two large vendor invoices remain unmatched pending PO confirmation. Headcount costs came in \$150K over budget due to a mid-month hire in customer success that wasn't in the original plan."*

Claim	Source	Period	Owner	Status	Gate Required
Cash increased 4%	Cash position report, treasury	Oct 2026	Treasury lead	Verified	None — if tied to bank statement

Increase driven by faster collections	Not identified — no AR aging/DSO cited	Oct 2026	AR/collections owner	Owner-needed	Pull AR aging comparison; confirm DSO change
AP aging improved slightly	Not identified — "slightly" undefined	Oct 2026	AP lead	Unsupported	Attach AP aging report with prior-period comparison
Two vendor invoices unmatched pending PO	Implied — AP exception log not attached	Oct 2026	AP lead / procurement	Inferred	Attach exception list with vendor names/amounts
Headcount costs \$150K over budget	Not identified — no payroll/budget file cited	Oct 2026	HR/FP&A	Owner-needed	Pull budget-to-actual headcount report
Overage caused by mid-month CS hire	Not identified — no hiring record referenced	Oct 2026	CS dept head	Owner-needed	Confirm hire date/comp against \$150K gap

Exercise 5 — "Just clean up the language" request

Before touching the wording: (1) What is each claim's source, and does the cleaned version still say only what the source supports? (2) Has anyone confirmed the causal claims, or are they still inferred/unsupported? (3) Is there a named approver reviewing substance, separate from whoever reviews prose?

Why this reproduces the fluency trap: editing for readability makes the note more fluent without closing the gap between fluent and grounded — it increases the exact property (surface confidence) that makes unverified claims dangerous, without touching evidence, sourcing, or ownership.

Exercise 6 — "AI output needs human review" is necessary but not sufficient

Necessary: puts at least one human between generation and the record. **Not sufficient:** "review" is undefined — a reviewer can check tone/formatting/grammar without ever opening the source file. **What would make it sufficient:** replace "review" with "clear the gate" — every claim must have a traceable source, every causal statement a named owner-confirmation, and a specific approver signing off on adequacy, not fluency.

Synthesis

Exercise 7 — Gate process for monthly variance commentary

Model input: period actuals-vs-budget export with explicit period boundaries/scope — no pipeline chatter. **Model output:** draft commentary + populated assessment artifact, delivered together. **Named approver:** FP&A manager/controller who did not draft the commentary. **Approval criteria:** (1) no unsupported claims allowed through; (2) every causal statement has a named owner-confirmation; (3) approver can articulate what would make the conclusion wrong.

Challenge

Exercise 8 — AS1105 evaluation of a generated variance note

Sufficient: fails by default — one narrative, no corroborating evidence. **Appropriate:** fails — pattern-matched from training data, not derived from this period's records. **Traceable:** fails — no citation, no source link. **Could AI meet AS1105?** Only if sourcing is structurally required at generation time (source-file/row attachment mandatory, causal language blocked without owner-confirmed driver, audit trail baked into the artifact) — not currently the case.

Exercise 9 — Operationalizing the adequacy bar across artifact types

Categories (increasing consequence): internal working notes → department reports → cross-functional decks → externally-facing/audit-relevant artifacts. **Calibration questions:** Who reads this and what will they do? What's the cost of being wrong? Could an external party later ask "how do you know this"? Does this become a source other artifacts cite? **Building it into the gate:** a written classification table mapping artifact type → required completeness → approver role, declared before generation begins, periodically audited.

Part 2 — Your Own Mycroft Project

This chapter's project applies the fluency-trap discipline to a real investment thesis: AI gathers and structures evidence; the human — Pranavi — makes every call.

Exercise 1 — When to Use AI

- Splitting a fluent thesis into individual claims — mechanical decomposition, checkable against the original text.
- Computing mechanical numbers off an options chain supplied by the user — arithmetic against pasted data, checkable against the chain.
- Summarizing a 10-K/10-Q or earnings-call transcript into candidate claims — extraction, verifiable line-by-line.

The tell: you're using AI correctly when you have independent criteria (a filing line, real options data) to judge the output.

Exercise 2 — When NOT to Use AI

- Deciding whether the thesis is true, or whether to buy — no source access, no stake in the outcome, real capital at risk.
- Judging whether an options signal is real institutional positioning or noise — needs actual term-structure data the model wasn't given.
- The buy/hold/sell decision itself — carries accountability that can't be delegated.

The tell: you've crossed the line when AI output becomes your reason to trade rather than an input to your own reasoning.

Exercise 3 — LLM Claim-Audit: theses/TITAN.md

Original thesis: *"Rising per-capita GDP and an expanding middle class in India are shifting spending toward branded, discretionary goods. Titan Company (India: TITAN) — India's largest branded jewelry and watch retailer — is positioned to benefit because branded jewelry/watches are a classic 'first discretionary upgrade' category as households cross from survival to lifestyle spending."*

#	Claim	Tag	Reason	Source / Evidence
1	India's per-capita GDP is rising	Verified (recent period), caveat	World Bank data confirms strong recent growth (~6–7%, peak ~8.6% post-2020); full 1961–2025 series is historically volatile, not a smooth trend	World Bank, GDP per capita growth (annual %) — India
2	India's middle class is expanding	Verified, sourcing caveat	6.3% annual growth (1995–2021), 31% of population currently middle class, cited by Finance Minister at a public forum	Forbes India, Jul 3 2026
3	Household spending shifting toward asset-building/durable goods	Verified	EAC-PM survey (HCES data, 2011-12 vs 2023-24) confirms shift from necessities toward durables (vehicles, appliances), incl. bottom 40% of households	Times of India / EAC-PM survey, Nov 2025

3a	This shift extends specifically to branded jewelry/watches	Taste / inferred — human decision	Survey names durable-goods categories (vehicles, fridges, phones) — does NOT mention jewelry or watches. Analytical bridge, not a sourced fact. Not elevated to verified, by Pranavi's own judgment.	No direct source found; flagged by human review
4	Jewelry/watches are a classic "first discretionary upgrade" category	Taste	Plausible, widely-repeated generalization; no isolating study found for the Indian market	None found
5	Titan is India's largest branded jewelry/watch retailer	Verified	~15M watches/year, 5th-largest watchmaker globally by volume, ~60% share of India's traditional watch market; Tanishq is India's top jewelry brand, >80% of Titan's sales	Hodinkee, Aug 2025
6	Titan is positioned to benefit from the trend	Mixed — real growth, real counter-variable	41% Q1 FY27 consumer-business growth confirmed (3rd consecutive >40% quarter). However, prior quarters (13%, 18% growth) were sharply suppressed by gold price spikes — a recurring, material driver independent of the middle-class narrative	Reuters, Jul 6 2026; Business Standard, multiple quarters

Human Judgment Notes

Claim 3a is the load-bearing gap in this thesis — the macro data and company data are each individually solid, but the connective claim (durable-asset spending → branded jewelry/watch spending) is not established by any source found. This is Pranavi's own explicit call, not a model-generated conclusion.

Gold price sensitivity is a live, material risk to claim 6 that the original one-sentence thesis did not mention.

Claim 2's middle-class figure is a secondhand citation of an OECD projection via a government minister's speech — directionally credible but not the primary source.

Final status: Verified — 1 (recent-period), 2 (sourcing caveat), 3, 5, 6 (partially). Held as taste/inferred by explicit human decision — 3a, 4.

Owner of this thesis: Pranavi

Exercise 5 — AI Validation Checklist Applied

Checklist item	Result	Note
Correctness	Pass	Nothing tagged "verified" without backing
Completeness	Pass	All distinct claims captured, including the connective assumption (3a)
Scope	Pass	No rating, target, or opinion added beyond what was in the source thesis
Signal vs. noise	N/A	No options-market claim present in this thesis
Owner test	Pass	Each needs-a-source claim names a specific document/data type
Failure-mode check	Flagged	Claim 3a initially looked "directionally supported" by the HCES survey; human review correctly caught that the survey's actual categories (durables) do not name jewelry/watches specifically

Verdict: Thesis is on firmer footing than an initial mismatched test case (Intel), but claim 3a remains a human judgment call, not a verified fact — correctly preserved as such in the final file rather than smoothed over.

Exercise 4 — CLI Repository Setup

Repository built via Claude Code inside the **chapter-1** project folder:

```
chapter-1/
├── book/
├── evidence/
├── signals/
├── theses/
├── TITAN.md
├── CANNOT-KNOW.md
├── CLAUDE.md
└── README.md
```

CLAUDE.md (verbatim, as required):

"The assistant never marks a claim 'verified', never labels an options signal 'real', and never recommends, sizes, or executes a trade. Only a human, citing a source, marks a claim verified. Only a human decides and places a trade. Read and analyze only — no brokerage write access, no order placement."

Inspection results:

Check	Result
README invented no claims/tickers	Pass — every statement traces to TITAN.md or CANNOT-KNOW.md
Files moved, not rewritten	Pass — TITAN.md content unchanged
CLAUDE.md matches rule verbatim	Pass
Nothing deleted	Pass

Part 3 — Completion Status

Exercise Set	Status
Conceptual Exercises 1–9 (Warm-up, Application, Synthesis, Challenge)	Complete
Your Own Mycroft — Exercise 1 (When to Use AI)	Complete
Your Own Mycroft — Exercise 2 (When NOT to Use AI)	Complete
Your Own Mycroft — Exercise 3 (LLM Claim-Audit → theses/TITAN.md)	Complete
Your Own Mycroft — Exercise 4 (CLI Repository Setup)	Complete
Your Own Mycroft — Exercise 5 (AI Validation Checklist)	Complete

Chapter 1 — The Fluency Trap — is fully complete.